

Low-Tech SOP Sensing over FTTH for Seismic Monitoring in Campi Flegrei: Benchmarking Against Interferometry and ϕ -OTDR

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Abstract We demonstrate, for the first time, PBS-based low-tech SOP sensing for seamless seismic monitoring using in-service FTTH infrastructure in the highly seismic area of Campi Flegrei. The approach detects earthquakes down to magnitude 1.9 and is positively benchmarked against consolidated interferometric phase and ϕ -OTDR techniques.

Introduction

Beyond data transmission, optical fibers are intrinsically sensitive to environmental perturbations: mechanical stresses induced by external disturbances modify both the light's optical phase [1] and the state of polarization (SOP) [2]. This dual sensitivity enables fiber networks to operate as distributed sensors over vast geographical areas. In highly seismic regions, continuous and extensive structural and environmental monitoring is crucial for risk mitigation and early response. However, conventional seismic sensing over fiber technologies like Distributed Acoustic sensing (DAS) [3, 4] often require dedicated infrastructure, specialized equipment, and significant deployment costs,

limiting scalability. To overcome these limitations, sensing techniques based on optical phase and SOP have gained momentum. The interferometric phase technique has been shown effective in detecting of seismic signals [1, 5] and compatible with existing traffic, although it requires expensive ultrastable source lasers. Seismic detection using SOP has been studied mostly on the oceanic scenario by extracting the full Stokes vectors set from coherent transceivers [6–8]. However, access to this data is typically prevented by proprietary digital signal processing (DSP) receivers or offered at insufficient sampling rates. In this work, we focus on a low-tech, low-cost and scalable approach leveraging a polarization beam splitter

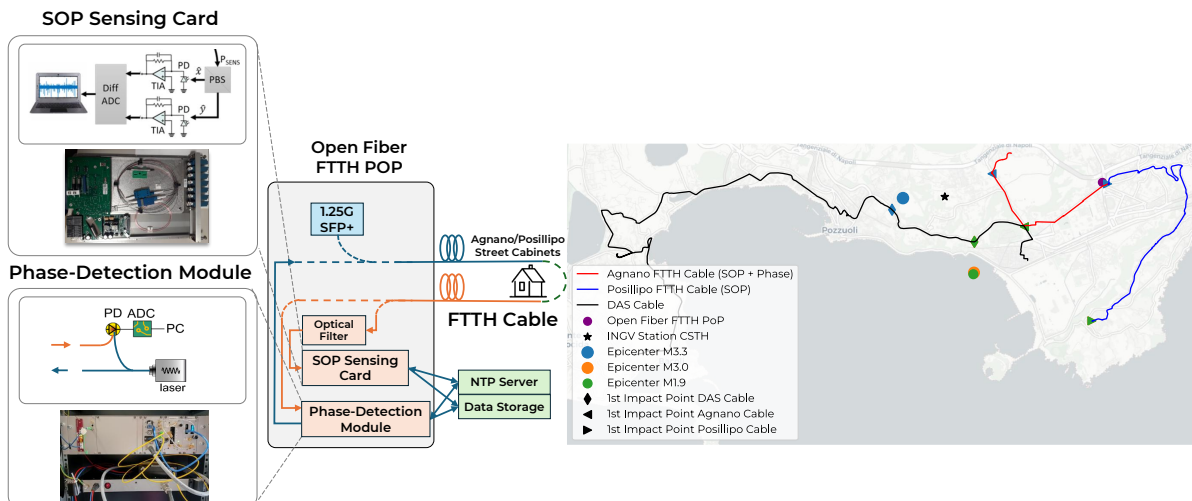


Fig. 1: Experimental setup of the FTTH link, sensing modules and map of deployments.

(PBS)-based device to monitor the SOP [9–11] to enable the seamless dual use of existing fiber-to-the-home (FTTH) infrastructure for earthquake monitoring in the terrestrial scenario. By tapping only a small portion of an in-service data channel, our approach enables pervasive seismic monitoring without impacting live traffic. We demonstrate, for the first time, that such scheme deployed over operational FTTH links can detect earthquakes down to magnitude 1.9. The proposed solution is benchmarked against established and more sophisticated fiber-sensing techniques, including interferometric phase sensing [1] and ϕ -OTDR (DAS) [4], showing strong agreement while significantly reducing system complexity and cost.

Experimental setup and Earthquake detection

The experimental testbed covers an area about 7 km wide in Campi Flegrei and is outlined in Fig.1. Most of the seismic activity in Campi Flegrei originates in the so-called *Solfatara di Pozzuoli*, an area of 700 meters in diameter placed in the middle of the covered area (also epicenter of the M3.0 event reported below) although many seismic events occur all around that area, even on the seafront. The DAS signal is propagated through a 22 km dark fiber in a production FTTH cable owned by the Italian operator TIM, co-located fibers in the same cable carrying live traffic [12]. The cable starts in the central office in Bagnoli (Naples), traverses the city of Pozzuoli and terminates in the city of Bacoli. The DAS interrogator is a Febus A1 using a gauge length of 10 meters a sampling frequency of 200 Hz and a spatial sampling of 4.8 meters resulting in 4587 spatial channels. The SOP and interferometric phase sensing techniques are instead deployed in a loop-back configuration over two cable segments carrying live data part of the Open Fiber (OF) FTTH network. Both segments depart in opposite directions from the Open Fiber Point of Presence (PoP) (purple marker in Fig.1), hosting also the optical signal sources and the acquisition devices, and terminate in street cabinets in the locations of Agnano (red cable) and Posillipo (blue cable) with lengths of 4.25 km and 7.25 km, respectively. Interferometric phase sensing is done only in the Agnano cable section, while SOP technique, being much cheaper, is deployed on both, enabling easier scalability. The block diagram Fig.1 shows the configuration of the Agnano branch utilizing both techniques, but generalizes to the Posillipo cable only for SOP sensing. Regarding the interferometric phase technique, at the OF PoP, the radiation from a commercial compact narrow-linewidth laser source is sent through the cable and looped back at the Agnano street cabinet using another fiber in the same cable. Back at the OF PoP the received signal is mixed with

the same reference laser source, then their interference is detected by a photodiode and digitized by an ADC to extract the phase, whose variation depends on the time-varying optical path caused by external disturbances. With regards to the SOP technique instead, at the OF PoP, a cheap commercial 1.25 Gbps intensity-modulated transceiver provided by SM-Optics is used as signal source. Being this inherently polarized we do not require expensive signals sources with this technique because the laser phase noise is identical among the orthogonal polarization components, thus enabling scalability. This signal is then 50/50 splitted and propagated through both Agnano and Posillipo cables using fibers carrying live data. The signal power launched in each fiber is around -2 dBm. While in previous field trials [13] the SOP was detected at the opposite cable end, in this case the 10G signal is looped back to the OF PoP at the street cabinets using another fiber in the same cable, as no space and power supply are available in the cabinets. Back at the OF PoP, each of the two cable signals is filtered and sent to the SOP monitoring device input developed by SM-Optics. However, the majority of the received power can be used to receive 10G channel data with no loss of sensing performance. The SOP sensing device developed by SM-Optics uses a simple approach based on a polarization beam splitter (PBS) to obtain the orthogonal polarization components [10], each digitized using a single photodiode, a trans-impedance amplifier (TIA) and an ADC with 100 Hz sampling rate. The two components are then normalized to the total instantaneous power and their difference is taken as the SOP signal, thus representing the Stokes vector S_1 in the receiver polarization frame. Although this technique is not able to capture the whole Poincaré sphere, the slow polarization drift allows to capture many events happening along the fiber whilst keeping costs low. Both interferometric phase and SOP signals are time-stamped and synchronized via an NTP server. The acquired data are sent daily to a remote virtual machine owned by Open Fiber for storage. Fig.2 showcases the capability of the low-tech SOP sensing to detect seismic events down to magnitude 1.9. The considered events are certified through the GOSSIP seismic catalog of INGV-Osservatorio Vesuviano (INGV-OV) [14], for which we report the unique database ID. SOP, phase, and INGV waveforms in Fig. 2 are shown after zero-phase 3rd-order Butterworth band-pass filtering (2–15 Hz for Fig. 2a and 1–25 Hz for Fig. 2b). Spectrograms are computed from the unfiltered traces using a 1 s Hamming window, 80% overlap and FFT length $8N$, while INGV traces are first compensated for instrument response to ground ve-

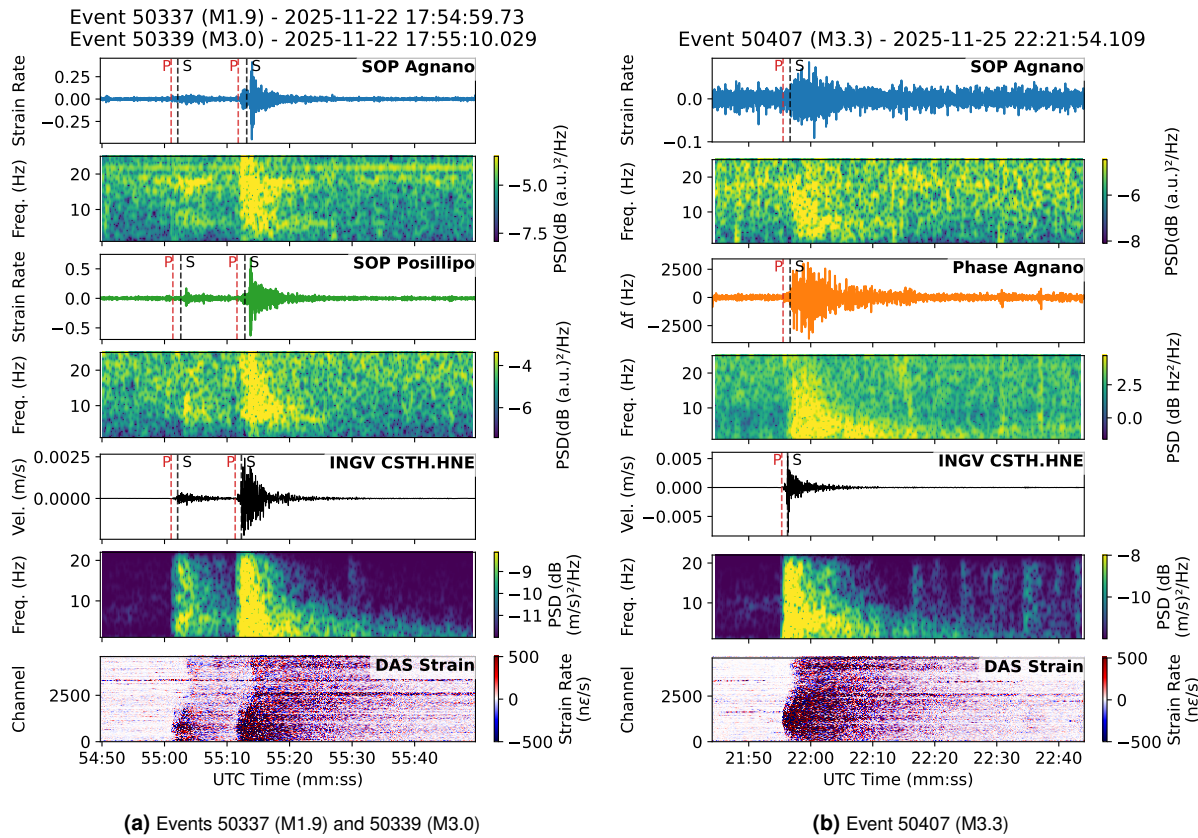


Fig. 2: Seismic events recorded by the multi-tech sensing testbed. (a) Sequence of two consecutive M1.9 and M3.0 events detected on 2025 Nov. 22nd using SOP, DAS and seismic stations with epicenter in the seafront of the phase/SOP testbed. (b) M3.3 detected on 2025 Nov. 25th detected by SOP, phase, DAS and seismic station with epicenter in *Solfatara di Pozzuoli*.

locity (VEL). Fig.2a show two consecutive events with epicenter in the seafront, M1.9 (id 50337) followed by a M3.0 (id 50339) after 10 seconds. We report the SOP measurements on both Agnano and Posillipo cables and the respective spectrograms. These measurements are compared to the DAS signals as a well-established, spatially resolved fiber-based technique and against the acquisition from a reference Csth seismic station, part of the INGV italian seismic network. Unfortunately, the interferometric phase system was offline during this period. In the SOP plots, we report with vertical lines the arrival times of the P and S phases computed by NonLinLoc code [15] using the velocity model at Campi Flegrei [16]. The location and origin time of the seismic events are taken from the GOSSIP seismic catalog of INGV-OV [14]. The nearest point to the earthquake epicenter is chosen as the seismic wave first impact point to the fiber and used to calculate the arrival time. Although noisier than seismic station data, the SOP data clearly show P and S waves of the larger magnitude event and the S wave of the M1.9 event, and the spectral signature is compatible with those expected by such seismic events. Fig.2b shows the detection of a

M3.3 event happened on 2025 Nov. 25 (id 50407) with epicenter in *Solfatara di Pozzuoli*. We report only the Agnano trace on the SOP side and compare also to the interferometric phase data as a well-established reference for fiber-based, space-integrated sensing established on the same fiber. The SOP and phase traces show consistent waveform features, and the estimated arrival times are compatible with the observed detections. We have also been able to detect using SOP and validate against well-established detection techniques also the M2.2 earthquake of 2025 Nov. 22nd (id 50334), not shown here due to limited space and we expect more successful detections as we progress in data post-processing.

Conclusions

We demonstrated a low-cost, PBS-based SOP sensing approach for seismic monitoring over in-service FTTH infrastructure. The system detects earthquakes down to magnitude 1.9 and shows strong agreement with interferometric phase sensing, ϕ -OTDR, and reference seismic stations. This simple and scalable solution enables pervasive seismic monitoring by seamlessly using existing telecom networks.

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